

1. Executive Summary

		Traffic Summary		Count	
Metric	Value	Total Flows Observed		86329	
Total Threats Mitigated	24	Total Packets Analyzed		5191528237	
High Priority	5	Malicious Packets		5162954449	
Low Priority	19	True Negatives Passed		71596	
Manual Releases	0				
Manual Blocks	0	Attack Vector		Count	
		ICMP Flood		0	
ML Metrics	Value	SYN Flood		15	
False Positives	43	UDP Flood		2	
FP Rate	0.05%	Uncertain		7	
IF Precision	99.75%				
IF Recall	92.90%	Action		Count	
IF F1-Score	96.21%	Quarantined		5	
IF Accuracy	98.66%	Time Ban		4	
		Blackhole		0	
		Released		6	

2. Performance Benchmark

2a. Isolation Forest: Anomaly Detection

Metric	Value	Description
Precision	99.75%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	92.90%	Share of actual attacks successfully flagged
F1-Score	96.21%	Balanced measure combining Precision and Recall
Accuracy	98.66%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.05%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	7.10%	Actual attacks that went undetected
True Positive Rate (TPR)	92.90%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.95%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 14639	FN 1118
Actual: Normal	FP 36	TN 70478

2b. Random Forest: Attack Classification

Metric	Value	Description
Precision	100.00%	Share of classified flows assigned the correct attack type
Recall	100.00%	Share of attacks of each type that were correctly identified
F1-Score	100.00%	Balanced measure combining Precision and Recall
Accuracy	100.00%	Overall proportion of correct classifications

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	14639	0	0
Act: ICMP	0	0	0
Act: UDP	0	0	0

2c. Response Latency

Metric	Value	Description
Detection Time	485.2 ms	Average time taken to detect an attack
Mitigation Response Time	5.8 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2d. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	1.34%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	4.52%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	5.83%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.19	3	3	2026-09-15 10:17:42	2026-09-15 10:19:03
10.0.0.16	2	2	2026-09-15 10:17:40	2026-09-15 10:18:50
10.0.0.17	2	2	2026-09-15 10:17:40	2026-09-15 10:18:50
10.0.0.18	2	2	2026-09-15 10:17:42	2026-09-15 10:18:52
10.0.0.6	1	1	2026-09-15 10:15:28	2026-09-15 10:16:29

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-15 10:06:39	10.0.0.17	DDoS	Uncertain	-	Low	Quarantined
2026-09-15 10:06:40	10.0.0.16	DDoS	Uncertain	-	Low	Quarantined
2026-09-15 10:06:48	10.0.0.18	DDoS	Uncertain	-	Low	Quarantined
2026-09-15 10:06:48	10.0.0.19	DDoS	Uncertain	-	Low	Quarantined
2026-09-15 10:06:49	10.0.0.17	DDoS	SYN Flood	91.8%	Critical	Time Ban (1m)
2026-09-15 10:06:50	10.0.0.17	DDoS	SYN Flood	91.8%	Critical	Time Ban
2026-09-15 10:06:50	10.0.0.16	DDoS	SYN Flood	93.3%	Critical	Time Ban (1m)
2026-09-15 10:06:50	10.0.0.16	DDoS	SYN Flood	93.3%	Critical	Time Ban
2026-09-15 10:06:58	10.0.0.18	DDoS	SYN Flood	90.9%	Critical	Time Ban (1m)
2026-09-15 10:06:58	10.0.0.19	DDoS	SYN Flood	90.9%	Critical	Time Ban (1m)
2026-09-15 10:06:58	10.0.0.18	DDoS	SYN Flood	90.9%	Critical	Time Ban
2026-09-15 10:07:00	10.0.0.19	DDoS	SYN Flood	91.8%	Critical	Time Ban
2026-09-15 10:07:50	10.0.0.17	DDoS	SYN Flood	91.8%	High	Released
2026-09-15 10:07:51	10.0.0.16	DDoS	SYN Flood	93.3%	Critical	Time Ban (2m)
2026-09-15 10:07:59	10.0.0.18	DDoS	SYN Flood	90.9%	High	Released
2026-09-15 10:07:59	10.0.0.19	DDoS	SYN Flood	91.8%	Critical	Time Ban (2m)
2026-09-15 10:08:00	10.0.0.17	DDoS	SYN Flood	93.2%	Critical	Time Ban (2m)
2026-09-15 10:08:09	10.0.0.18	DDoS	SYN Flood	91.4%	Critical	Time Ban (2m)
2026-09-15 10:13:28	10.0.0.6	DDoS	UDP Flood	92.4%	High	Time Ban (1m)
2026-09-15 10:13:34	10.0.0.5	DDoS	Uncertain	59.1%	Low	Released
2026-09-15 10:13:39	10.0.0.1	DDoS	Uncertain	58.9%	Low	Released
2026-09-15 10:14:29	10.0.0.6	DDoS	UDP Flood	92.4%	High	Released
2026-09-15 10:14:58	10.0.0.6	DDoS	Uncertain	48.4%	Low	Quarantined
2026-09-15 10:18:52	10.0.0.19	DDoS	SYN Flood	93.4%	High	Released

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.19	SYN Flood	0.6537	93.4%	Critical	Phase 2	3	10s	2026-09-15 10:19:03	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.5775	93.3%	High	Phase 2	2	1m 10s	2026-09-15 10:18:52	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.5913	93.4%	High	Phase 2	1	1m 10s	2026-09-15 10:18:52	Ban Expired
10.0.0.16	SYN Flood	0.6387	92.5%	Critical	Phase 2	2	1m 10s	2026-09-15 10:18:50	Attack persisted - escalating to time ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.17	SYN Flood	0.6430	92.1%	Critical	Phase 2	2	1m 10s	2026-09-15 10:18:50	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6471	90.9%	Critical	Phase 2	1	10s	2026-09-15 10:17:52	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6521	91.7%	Critical	Phase 2	1	10s	2026-09-15 10:17:52	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6363	92.5%	Critical	Phase 2	1	10s	2026-09-15 10:17:50	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6423	92.1%	Critical	Phase 2	1	10s	2026-09-15 10:17:50	Attack persisted - escalating to time ban
10.0.0.6	UDP Flood	0.5891	92.4%	High	Phase 2	1	1m 1s	2026-09-15 10:16:29	Ban Expired

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE